Azure Event Hubs Knowledge Base

Page 1 — Azure Event Hubs Overview

Azure Event Hubs is a fully managed, real-time data streaming platform designed to ingest and process large volumes of events. It is commonly used for telemetry, application logs, clickstreams, IoT data, and other high-throughput event streams.

RAG Learning Context

This document is intentionally created as a sample enterprise knowledge base for learning RAG data ingestion with LangChain. It will later be reused for document processing, chunking, embedding, indexing, and retrieval exercises.

Page 2 — Event Hubs Core Concepts

An Event Hubs namespace is a management container for one or more Event Hubs. An Event Hub is a logical endpoint where producers send events and consumers read events.

Events are records representing information generated by applications, devices, services, or other systems. Event Hubs is designed for high-throughput ingestion.

Page 3 — Partitions and Consumer Groups

A partition is an ordered sequence of events within an Event Hub. Partitions allow event ingestion and consumption to scale by distributing work across multiple parallel streams.

Consumer groups provide independent views of an event stream. Multiple applications can consume the same Event Hub independently by using separate consumer groups.

Page 4 — Event Producers and Consumers

Producers publish events to Event Hubs. Producers can include applications, IoT devices, services, or data platforms that generate telemetry and operational data.

Consumers read events from Event Hubs and process them using services such as Azure Functions, Azure Stream Analytics, Azure Databricks, or custom applications.

Page 5 — Typical Azure Event Hubs Architecture

A common architecture is: event producers generate telemetry or application events; Azure Event Hubs receives and buffers the event stream; consumer applications process the events; and downstream systems store, analyze, or visualize the resulting data.

For RAG learning, this PDF provides multiple pages so that we can observe how a PDF loader represents page-level content and metadata as LangChain Document objects.